

|     |     |     |     |     |     |       |
|-----|-----|-----|-----|-----|-----|-------|
| 1   | 2   | 3   | 4   | 5   | 6   | Total |
| /20 | /20 | /20 | /20 | /20 | /20 | /100  |

Name: \_\_\_\_\_

June 23, 2019. Sunday

Department/School: \_\_\_\_\_

SID: \_\_\_\_\_

***University Physics***

**Final Examination**

**Kuang Yaming Honors School, Nanjing University**

$$\Gamma(\nu) = \int_0^{+\infty} e^{-x} x^{\nu-1} dx, \quad \Gamma(\nu+1) = \nu\Gamma(\nu), \quad \Gamma(1) = 1, \quad \Gamma(1/2) = \sqrt{\pi}.$$

Select five out of following six problems.

1.(20pts.) There happen two events that are at the same place and a time interval  $\Delta t' = 300\text{s}$  apart in the  $S'$  reference frame. In the  $S$  reference frame, the two events are  $\Delta t = 500\text{s}$  apart. The  $S'$  frame moves with speed  $u$  relative to  $S$  frame along the common  $x$  and  $x'$  direction.

(a) What is the speed  $u$  ?

(b) What is the space interval  $d$  between two events in the  $S$  frame?

2. (20pts.) A block of ice of refraction index  $n$  floats on the water of a river and travels with speed  $v$ . A beam of light propagates through the transparent ice block in the same direction of the velocity of the moving ice.

(a) What is the speed of the light  $v_1$  in the ice to the observer that travels with the ice block?

(b) What is the speed of the light  $v_2$  in the ice to the observer standing on the shore of the river?

3.(20pts.) A block of ice of mass  $2\text{ kg}$  and at temperature of  $0^\circ\text{C}$  is placed into a bath of water of  $10\text{ kg}$  at temperature  $10^\circ\text{C}$ . The ice and the bath of water as a whole are adiabatically enveloped. After some time, the system is in equilibrium again. The latent heat of melting of ice (at  $0^\circ\text{C}$ ) is  $3.35 \times 10^5\text{ J/kg}$  and the specific heat capacity of water at constant pressure is  $4.187 \times 10^3\text{ J/kg} \cdot \text{K}$ .

- (a) Find  $\Delta S_1$ , the change in entropy of the ice .
- (b) Find  $\Delta S_2$ , the change in entropy of the bath of water.
- (c) Find  $\Delta S_t$ , the change in entropy of the total system.

4. (20pts.) A system of gas of fixed number of molecules is with two thermodynamic degrees of freedom (or two independent thermodynamic coordinates) only. The volume  $V$  and the internal energy  $U$  can be chosen as the two thermodynamic degrees of freedom (or thermodynamic coordinates).

(a) Write down the differential form of the temperature  $T$  with respects of volume  $V$  and internal energy  $U$ .

(b) Show from (a) that 
$$dU = \left[ \left( \frac{\partial T}{\partial U} \right)_V \right]^{-1} dT - \left[ \left( \frac{\partial T}{\partial U} \right)_V \right]^{-1} \left( \frac{\partial T}{\partial V} \right)_U dV.$$

(c) Show that, if it were found in the Joule's experiment of free expansion of the gas that  $\left( \frac{\partial T}{\partial V} \right)_U = 0$ , the internal energy  $U$  of the gas could be the function of temperature  $T$  only.

(d) We all know that the internal energy of the ideal gas is the function of temperature  $T$  only. Then, if a mole of ideal gas freely expands from  $v_1$  to  $v_2$  as in the Joule's experiment, what is the change in entropy  $\Delta S$  for the gas?

5. (20 pts.) Let  $U$ ,  $F$ ,  $S$ ,  $T$ ,  $p$ ,  $V$  and  $C_V$  denote the internal energy, Helmholtz free energy, entropy, temperature, pressure, volume, and heat capacity at constant volume of a gas of fixed particles, respectively.

(a) Show that  $TdS = C_V dT - T \left( \partial^2 F / \partial T \partial V \right) dV$ .

(b) Show that  $(\partial V / \partial T)_S = -(C_V / T) (\partial T / \partial p)_V$ .

(c) Show that  $(\partial U / \partial p)_V = -T (\partial V / \partial T)_S$ .

6. (20pts.) From the kinetic theory of gases, the pressure of a gas can be determined by the average of the quantity  $nmv^2/3$ , where  $n$  is the number density of molecules,  $m$  is the mass of the molecule and  $v$  is the speed of the molecule.

(a) Find the pressure of the gas in terms of temperature  $T$  by applying the Maxwell distribution.

(b) Find the fluctuation (or uncertainty) of the pressure  $\Delta p$ . (  $\Delta p = \sqrt{\langle (nmv^2/3)^2 \rangle - \langle nmv^2/3 \rangle^2}$  )